THE VIRTUAL MEET EXPERIENCE

2017 - 2018

HS VIRTUAL CHALLENGE MEET #2

COMPUTER SCIENCE

DO NOT OPEN TEST UNTIL TOLD TO DO SO

2017-2018 Virtual Challenge Meet #2 – Computer Science

Note: Correct responses are based on Java, **J2sdk v 1.7.25**, from Sun Microsystems, Inc. All provided code segments are intended to be syntactically correct, unless otherwise stated (i. e. `error` is an answer choice) and any necessary Java 2 Standard Packages have been imported. Ignore any typographical errors and assume any undefined variables are defined as used. **For all output statements, assume that the System class has been statically imported…** ***import static java.lang.System.*;***

QUESTION 1

What is ABC_{16} - 212_{16}?

A. 100010101010_2 B. 2218_{10} C. 4252_8 D. $8AA_{16}$ E. All are equivalent.

QUESTION 2

What is output by the code to the right?

A. 2 B. 0

C. 2.0 D. 0.0

E. There is no output due to a compile error.

```
int g=7;
double v=g/3;
out.println(v);
```

QUESTION 3

What is output by the code to the right?

A. 9 dunzo B. dunzo9

C. 9dunzo D. dunzo 9

E. There is no output due to an error.

```
int y=9;
String fun="dunzo";
out.printf("%.2f%s", y,fun);
```

QUESTION 4

What is output by the code to the right?

A. ive is > seven B. e is > seven

C. 212ive is > seven D. ve is > seven

E. There is no output due to a syntax error.

```
String g="five is > seven";
g.replaceAll("f", "212");
System.out.println(g.substring(3));
```

QUESTION 5

What is output by the code to the right?

A. true true B. false true

C. false false D. true true

E. There is no output due to a syntax error.

```
boolean t=true;
boolean f=false;
out.print(t || f);
out.print(" ");
out.print(f ^ t);
```

QUESTION 6

What is output by the code to the right?

A. 10 B. 9

C. 10.0 D. 9.0

E. There is no output due to an error.

```
int h=0;
h+=Math.ceil(9.6);
out.print(h);
```

QUESTION 7

What is the output by the code to the right?

A. 3212 B. 2

C. 212 D. 32

E. 321

```
int y=10, x=9;
if(y<212 || ++x!=10) {
 y=212;
 out.print(3);
}
if(x+y==222)
 out.print(21);
out.print(2);
```

QUESTION 8

What is the output by the code to the right?

A. 16 B. 17

C. 18 D. 19

E. 20

```
int k=212, v=0, num=0;
for(;v<k;v+=4) {
 k-=9;
 num++;
}
out.print(num);
```

QUESTION 9

What is the output by the code to the right ?

A. 1029 B. 517

C. 261 D. 133

E. There is no output due to a syntax error.

```
String k="*";
for(int y=0;y<8;y++)
 k+=k;
for(int y=0;y<5;y++)
 k+="*";
out.print(k.length());
```

QUESTION 10

What is the output by the code to the right ?

A. 4 B. 8

C. There is no output due to a syntax error

D. There is no output due to a runtime error

E. Output cannot be determined until runtime.

```
int[][]mat=new int[][]
           {{1,2,3},{4,5,6},{7,8,9}};
out.print(mat[2][1]);
```

QUESTION 11

Which of the following could read in the following number so it could be used in a calculation? **891123**

A. `file.nextInt()` B. `file.nextDouble()`

C. `file.nextLine()` D. `file.next()`

E. More than one of these.

QUESTION 12

What is the output by the code to the right ?

A. 729 B. 243

C. 2187 D. 64

E. There is no output due to a syntax error.

```
int y=1;
for(int y=0;y<7;y++)
  y*=3;
out.print(y);
```

QUESTION 13

What is the correct order of precedence for the operators to the right ?

A. III, IV, II, I B. IV, III, II, I

C. I, II, III, IV D. II, IV, III, I

E. IV, III, II, I

```
I. []
II. !
III. casting
IV. &
```

QUESTION 14

What is the output by the code to the right ?

A. 4 B. 64

C. 8 D. 32

E. 16

```
out.print(Character.SIZE);
```

QUESTION 15

Which could not replace **<1*>** in the code to the right?

A. `Object`
B. `List<Integer>`
C. `Collection<Integer>`
D. `ArrayList<Integer>`
E. All of the above could replace **<1*>**

```
<1*> a=new ArrayList<Integer>();
```

QUESTION 16

What is the output by the code to the right ?

A. `0`
B. `127`
C. `63`
D. `17`
E. There is no output due to a syntax error.

```
out.print(63 & 127);
```

QUESTION 17

What is the output by the code to the right ?

A. `198`
B. `135`
C. `165`
D. `0`
E. There is no output due to an error.

```
int count=0;
for(int y=0;y<10;y++)
 for(int f=0;f<y;f++)
    count+=3;
out.print(count);
```

QUESTION 18

What is output by the code to the right?

A. `[3, 7, 212, 9, 8]`
B. `[3, 7, 8, 9, 212]`
C. `[3, 7, 212, 8, 9]`
D. There is no output due to a runtime error.
E. There is no output due to a syntax error.

QUESTION 19

Which data structure is implemented by the `PriorityQueue` class ?

A. Queue
B. Stack
C. Binary Search Tree
D. Min-heap
E. HashTable

```
PriorityQueue q;
q=new PriorityQueue<Integer>();
q.add(8);
q.add(9);
q.add(212);
q.add(7);
q.add(3);
out.print(q);
```

QUESTION 20

What is the big O runtime of adding a value to a hashset?

A. O(n)
B. O(n^2)
C. O(n !)
D. O(1)
E. None of the above.

QUESTION 21

What is output by the code to the right?

A. `129`
B. `558d`
C. `28b`
D. There is no output due to a compile error.
E. There is no output due to a runtime error.

```
int y;
y=Integer.parseInt("100101001",2);
String s=Integer.toString(y,6);
y=Integer.parseInt(s,8);
s=Integer.toBinaryString(y);
y=Integer.parseInt(s,3);
s=Integer.toString(y,16);
out.println(s);
```

QUESTION 22

Which of the following could replace **<1*>** in the code to the right?

A. `super`
B. `this`
C. `self`
D. A and B
E. A and C

QUESTION 23

Which of the following could replace **<2*>** in the code to the right?

A. `A`
B. `Object`
C. `B`
D. All of the above.
E. A and C

QUESTION 24

Assuming that **<1*>** and **<2*>** are filled correctly, what is the output by the following client code ?

```
A a=new A(7);
out.println(a);
```

A. 7
B. A
C. There is no output due to a syntax error.
D. Output cannot be determined until runtime.
E. This is no output due to a runtime error.

QUESTION 25

Assuming that **<1*>** and **<2*>** are filled correctly, what is the output by the client code to the right?

A. 000
B. 02000
C. 32333
D. Output cannot be determined until runtime
E. This is no output due to a runtime error.

```
class A{
  private int a;
  public A(int a){
     <1*>.a=a;
  }
  public A(){
     out();
     out.print(2);
  }
  public void out(){
     out.print(a);
  }
}
class B extends A{
  private int a;
  public B(){
     a=3;
     fun();
  }
  public void fun(){
     super.out();
  }
}
/////////////client code////////////
<2*> b=new B();
b.out();
b.fun();
```

QUESTION 26

What is the sum of the ASCII values of all the characters in String `k` to the right?

A. 1725
B. 1746
C. 1614
D. 1704
E. None of the above.

```
String k="212 is greater than 7";
```

QUESTION 27

Which of the following is not a valid java identifier?

A. `Rahul1sBad`

B. `__212__`

C. `Mr212`

D. `cypress woods`

E. All of the above are valid.

QUESTION 28

What is output by the line marked `//q28`?

A. `146`

B. `422`

C. `147`

D. `421`

E. `0`

QUESTION 29

What is output by the line marked `//q29`?

A. `0`

B. `13`

C. `12`

D. `10`

E. `11`

QUESTION 30

What is output by the line marked `//q30`?

A. `50`

B. `49`

C. `72`

D. `71`

E. `0`

```
public int recur(int s){
  if(s>=10)
     return s;
  return recur(s+1)+recur(s+2)+(s+1);
}


///////client code//////////////////
out.println(recur(6)); //q28
out.println(recur(12)); //q29
out.println(recur(8)); //q30
```

QUESTION 31

What is the big O runtime of the code to the right, N=`a.length`?

A. O(logN)
B. O(N)
C. O(N*N)
D. O(1)
E. O(NlogN)

QUESTION 32

What is the output by the code to the right?

A. `[212, 9, 7, 7, 4, 2, 1, -8, 3]`
B. `[212, 9, 7, 7, 4, 3, 2, 1, -8]`
C. `[-8, 1, 2, 3, 9, 7, 212, 4, 7]`
D. `[-8, 1, 2, 3, 4, 7, 7, 9, 212]`
E. Output cannot be determined until runtime.

QUESTION 33

Which sorting algorithm is implemented in the code to the right?

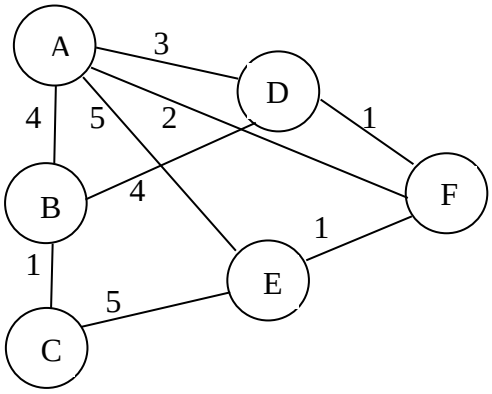
A. Merge Sort
B. Insertion Sort
C. Bubble Sort
D. Selection Sort
E. Stooge Sort

```
int[] a=
{1, 4, 3, 7, 9, 2, 212, -8, 7};
for(int ou=0;ou<a.length;ou++)
{
  int curr=ou;
  for(int in=ou+1;in<a.length;in++)
  {
     if(a[curr]<a[in])
        curr=in;
  }
  if(ou==4)
     out.println(Arrays.toString(a));
  int temp=a[curr];
  a[curr]=a[ou];
  a[ou]=temp;
}
```

QUESTION 34

What is the cost of the cheapest cost path from node A to E in the graph to the right?

A. 3
B. 5
C. 10
D. 2
E. 212



QUESTION 35

What kind of graph is represented to the right ?

A. undirected unweighted
B. connected weighted directed
C. weighted undirected
D. unweighted directed
E. connected weighted undirected

QUESTION 36

What is the most simplified equivalent to the truth table to the right ?

A. `A^B^C`
B. `A`
C. `A&B|C`
D. `A|B&C`
E. `C`

A	B	C	
1	1	1	1
1	1	0	1
1	0	1	1
1	0	0	0
0	1	1	1
0	1	0	0
0	0	1	1
0	0	0	0

QUESTION 37

What could replace **<1*>** in the code to the right?

A. `Object`
B. `List`
C. `ArrayList`
D. C and B
E. All of the above.

QUESTION 38

What is output by the line marked `//1` in the code to the right?

A. `286`
B. `6`
C. `290`
D. `10`
E. There is no output due to an error.

QUESTION 39

What is output by the line marked `//2` in the code to the right?

A. `JFK ABC 4 3 900 DOGH 212`
B. `212 DOGH 900 4 3 ABC JFK`
C. `4 3 212 900 ABC JFK DOGH`
D. `DOGH JFK ABC 900 212 4 3`
E. There is no output due to an error.

QUESTION 40

Which data steucture is implemented in class `Structure` to the right?

A. Min-heap
B. Hashtable
C. Array based list
D. Linked list
E. Stack

```
class Structure{

  <1*>[] in;
  int size;

  public Structure(int size){
    this.size=size;
    in=new ArrayList[size];
    for(int y=0;y<size;y++)
      in[y]=new ArrayList<String>();
  }

  public void add(String k){
    int pos=val(k)%size;
    in[pos].add(k);
  }

  public String toString(){
    String s="";
    for(ArrayList<String> k:in){
      for(String g:k)
          s+=g+" ";
    }
    return s;
  }

  public static int val(String k){
    int tot=0;
    for(char c:k.toCharArray()){
      tot+=(int)c;
    }
    return tot;
  }

}

//////////////////////////////////////
////////client code/////////////////
Structure s=new Structure(20);
out.println
         (Structure.val("DOGH")); //1
s.add("212");
s.add("ABC");
s.add("JFK");
s.add("DOGH");
s.add("900");
s.add("4 3");
out.println(s); //2
```

Name _____ Final Score _____ _____

School _____ 2nd Grading _____ _____

Division: 1A 2A 3A 4A 5A 6A Grade: 9 10 11 12 1st Grading _____ _____
score initials

The Virtual Challenge Meets
Computer Science - Student Answer Sheet

1)	_____	21)	_____
2)	_____	22)	_____
3)	_____	23)	_____
4)	_____	24)	_____
5)	_____	25)	_____
6)	_____	26)	_____
7)	_____	27)	_____
8)	_____	28)	_____
9)	_____	29)	_____
10)	_____	30)	_____
11)	_____	31)	_____
12)	_____	32)	_____
13)	_____	33)	_____
14)	_____	34)	_____
15)	_____	35)	_____
16)	_____	36)	_____
17)	_____	37)	_____
18)	_____	38)	_____
19)	_____	39)	_____
20)	_____	40)	_____

Total Number Correct _____ x 6 = _____

Total Number Incorrect _____ x -2 = _____

Final Score (record here and top right) _____

2017-2018 Virtual Challenge Meet #2
Computer Science - KEY

1) E
2) C
3) E
4) B
5) A
6) A
7) D
8) B
9) C
10) B

11) E
12) E
13) C
14) E
15) E
16) C
17) B
18) A
19) D
20) D

21) B
22) B
23) C
24) D
25) B
26) A
27) D
28) A
29) C
30) A

31) C
32) A
33) D
34) A
35) E
36) C
37) C
38) C
39) B
40) B

Note to Graders:

- All provided code segments are intended to be syntactically correct, unless otherwise stated (e.g. error is an answer). **Ignore any typographical errors**.
- Any necessary Standard Java 2 Packages are assumed to have been imported as needed.
- Assume any undefined (undeclared) variables have been defined as used.

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